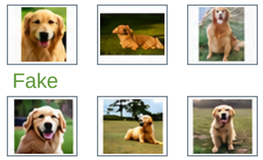
Support set S

K real + K fake

Real



Fake

Support preparation · before inference

Image Encoder
CLIP · L12

Cached features

z_{ci}, μ_c μ_0 μ_1

Support-only prompt fitting

CLIP → CE ← y t_0, t_1

P^*

Real · Fake · Frozen · Prompt · Forward · Fitted prompt dependency

Query inference

Query set Q



$x \in Q$

Image Encoder
CLIP · L12

q

Query-Conditioned Prototype Mixing

A

$\tilde{\mu}_c(q)$

p_p

Image Encoder
CLIP · L24, prompted

$v(x)$

Stage-wise Visual Prompting

Cosine margin

Fixed text anchors
t_0 t_1

p_v

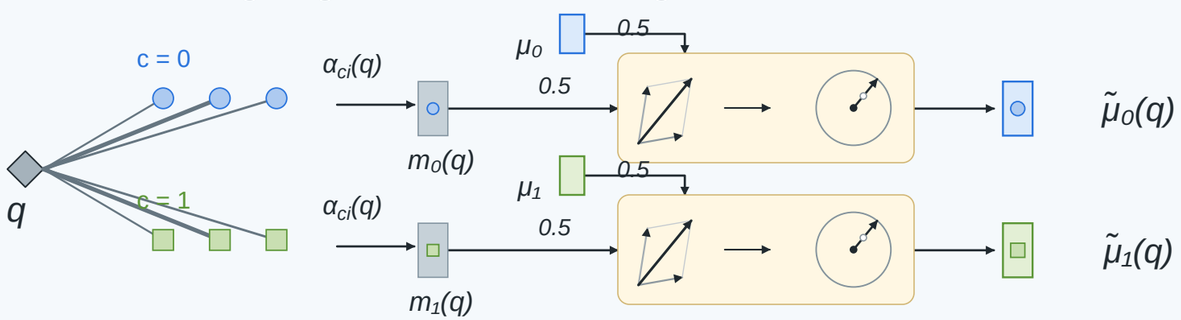
0.5 · 0.5

Real / Fake

$p > 0.5$: fake

$p = (p_p + p_v)/2$

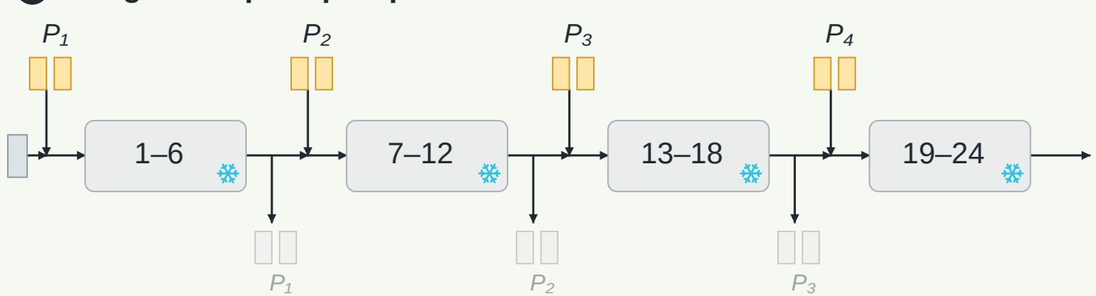
A Class-wise weighting and directional mixing



$s_p = 15\, q^T(\tilde{\mu}_1(q) - \tilde{\mu}_0(q))$, $p_p = \sigma(s_p)$

$\alpha_{ci}(q) = \text{softmax}_i(15\, q^T z_{ci})$

B Stage-wise prompt replacement



CLS + image tokens continue

$4 \times 2 \times 1024 = 8{,}192$

$p_v = \sigma(15\, v(x)^T(t_1 - t_0))$